

Teacher Notes

This section provides teachers with a quick reference for each chapter: the intended learning outcomes, common misconceptions to watch for, suggested classroom activities, and remediation tips for learners who need extra support.

Chapter 1: We the Travellers - I

Learning Outcomes: Students can read, write, compare, order and round large numbers (up to 6 digits) using real travel-based data.

Common Misconceptions: Students often compare numbers by counting digits only without checking place value carefully; some confuse the comma placement in the Indian number system.

Suggested Classroom Activities: Use real bus/train ticket samples or newspaper distance tables for students to read aloud and compare numbers.

Remediation Tips: Use place-value charts and abacus/base-ten blocks; practise reading numbers aloud digit-group by digit-group before writing them in words.

Chapter 2: Fractions

Learning Outcomes: Students can represent, compare and add/subtract like fractions, and apply fractions to real sharing situations.

Common Misconceptions: Believing a larger denominator always means a larger fraction; forgetting that fraction parts must be equal in size.

Suggested Classroom Activities: Paper-folding activities, sharing snacks or toffees equally in class, drawing shaded shapes.

Remediation Tips: Use physical objects (rotis, paper strips, fruit) before moving to symbols; reinforce “equal parts” with hands-on cutting.

Chapter 3: Angles as Turns

Learning Outcomes: Students can identify angle types as turns and relate them to clocks and directions.

Common Misconceptions: Confusing obtuse angles with reflex angles; assuming a visually ‘bigger looking’ angle is always the larger turn without checking carefully.

Suggested Classroom Activities: Use real clocks and body-turning games (face North, turn quarter/half turns) to build a physical sense of angles.

Remediation Tips: Use clock models and direction charts (N/S/E/W) for hands-on, kinesthetic practice before working with numbers alone.

Chapter 4: We the Travellers - II

Learning Outcomes: Students can calculate time durations, convert time units, and apply the basic speed-distance-time relationship.

Common Misconceptions: Treating time as a base-10 system (60 vs 100) when adding/subtracting; confusing am and pm.

Suggested Classroom Activities: Create a classroom timetable; use toy clocks for duration practice; analyse real bus/train timetables.

Remediation Tips: Practise time addition/subtraction using a number line; repeat conversions between hours and minutes frequently.

Chapter 5: Far and Near

Learning Outcomes: Students can convert between length units and estimate or compare real-life distances.

Common Misconceptions: Errors in decimal-point placement during unit conversion; confusion about which unit is larger.

Suggested Classroom Activities: Measure classroom objects with rulers/tape; play estimation games comparing guessed and actual lengths.

Remediation Tips: Use a metric conversion ladder/chart and lots of hands-on measuring with real tools.

Chapter 6: The Dairy Farm

Learning Outcomes: Students can convert and compute with capacity units, and solve real-world dairy/farm word problems.

Common Misconceptions: Forgetting to convert litres and millilitres to the same unit before adding or subtracting.

Suggested Classroom Activities: Use empty bottles of different capacities for hands-on pouring and comparison activities.

Remediation Tips: Practise with a unit-conversion chart; build up from simple addition problems to multi-step ones.

Chapter 7: Shapes and Patterns

Learning Outcomes: Students can classify 2D/3D shapes by properties, extend patterns, and calculate perimeter.

Common Misconceptions: Confusing perimeter with area; mixing up faces, edges and vertices of 3D shapes.

Suggested Classroom Activities: Build 3D shapes with clay or paper; create and extend patterns using classroom objects.

Remediation Tips: Use physical shape models; count edges/vertices by touching real objects before abstract questions.

Chapter 8: Weight and Capacity

Learning Outcomes: Students can convert and operate with weight units, and apply this to shopping/market situations.

Common Misconceptions: Forgetting to convert kg and g to the same unit before adding or subtracting, similar to capacity errors.

Suggested Classroom Activities: Bring empty packets with weight labels to class; do a weighing activity with a classroom scale.

Remediation Tips: Repeated practice with a conversion table; use real packaging examples for context.

Chapter 9: Coconut Farm

Learning Outcomes: Students can apply multiplication/division to grouping and farm-context problems, and calculate simple area and profit.

Common Misconceptions: Confusing when to multiply versus divide in word problems.

Suggested Classroom Activities: Role-play a farm market with buying and selling; use grid paper to visualise area.

Remediation Tips: Use repeated addition to bridge to multiplication; use concrete grouping activities with counters or seeds.

Chapter 10: Symmetrical Designs

Learning Outcomes: Students can identify and create line symmetry, and connect it to cultural art forms.

Common Misconceptions: Confusing rotational symmetry with line symmetry; assuming every shape is symmetric.

Suggested Classroom Activities: Paper folding and cutting; rangoli/mehendi design activities in class.

Remediation Tips: Use small mirrors to check symmetry; start with simple shapes before complex designs.

Chapter 11: Grandmother's Quilt

Learning Outcomes: Students can relate fractions to parts of shapes, calculate area of square/rectangular patches, and recognise repeating patterns.

Common Misconceptions: Assuming area scales the same way as side length; errors

in fraction-of-a-shape reasoning.

Suggested Classroom Activities: Paper-cutting square/triangle patchwork activity; grid-based quilt design exercises.

Remediation Tips: Use grid paper and physical cut-outs to reinforce area and fraction concepts together.

Chapter 12: Racing Seconds

Learning Outcomes: Students can convert between seconds and minutes, and compare/calculate time differences in sports contexts.

Common Misconceptions: Treating seconds and minutes as a base-10 system (60 vs 100).

Suggested Classroom Activities: Stopwatch races in PE class; recording and comparing times on a chart.

Remediation Tips: Use a number line to show the 60-second wraparound; practise frequent small conversions.

Chapter 13: Animal Jumps

Learning Outcomes: Students can compare and order decimal distances, and perform decimal addition/subtraction/multiplication in measurement contexts.

Common Misconceptions: Misreading decimal place value (e.g., thinking $0.45 > 0.6$); errors in aligning decimal points when adding.

Suggested Classroom Activities: Long jump measurement activity in PE; decimal number-line practice.

Remediation Tips: Use base-ten blocks or decimal grids; align decimal points using grid paper.

Chapter 14: Maps and Locations

Learning Outcomes: Students can use cardinal directions and interpret/apply map scale to find real distances.

Common Misconceptions: Confusing map distance with actual distance; errors in direction (clockwise vs anticlockwise turns).

Suggested Classroom Activities: Create a simple classroom or school map with scale; play direction-following games in the playground.

Remediation Tips: Use a compass or simple direction chart; practise scale conversion with concrete map examples.

Chapter 15: Data Through Pictures

Learning Outcomes: Students can read and interpret pictographs/bar graphs, calculate totals and comparisons, and construct simple graphs from given data.

Common Misconceptions: Forgetting to multiply by the key value in pictographs; misreading the scale on a bar graph's axis.

Suggested Classroom Activities: Class survey and graph-making activity; a 'news-paper data hunt' for real graphs.

Remediation Tips: Practise reading the key/scale first before answering any question; use grid paper for constructing bar graphs.

Answer Key

Answers to all worksheets are provided below, in chapter order. For open-ended reflection and real-life application questions, and for any drawing/geometry activity, teachers should use their own judgement: these are marked as “Teacher Verification Required.”

Worksheet 1: We the Travellers - I – Answer Key

1. Forty-five thousand six hundred seventy-eight
2. 62,409
3. 7,000
4. 5214, 5241, 5412, 5421
5. 38,902
6. 1,188 km
7. Rs. 48,500
8. 6,700
9. 24,000
10. 10,379
11. 1,070
12. $8,641 - 1,468 = 7,173$
13. $30,000 + 700 + 6$
14. $91,432 > 91,423$
15. 45,913 km
16. Digit sum of 16,089 = $1+6+0+8+9 = 24$; $24 \times 1,000 + 200 = 24,200$ tickets
17. Estimate: $8,000+9,000+8,000=25,000$; Actual: 25,610
18. Smallest 5,550; Largest 5,649
19. Rs. 95,003
20. 345 km 670 m

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 2: Fractions – Answer Key

1. $\frac{2}{5}$
2. $\frac{3}{7}$
3. $\frac{3}{5}$
4. $\frac{5}{9}$
5. Numerator 7, denominator 12
6. $\frac{6}{10} = \frac{3}{5}$
7. $\frac{12}{30} = \frac{2}{5}$
8. $\frac{4}{6} = \frac{2}{3}$
9. $\frac{6}{9} = \frac{2}{3}$
10. $\frac{4}{10} = \frac{2}{5}$
11. Used $\frac{8}{12} = \frac{2}{3}$; remains $\frac{4}{12} = \frac{1}{3}$
12. $\frac{6}{9}$
13. $\frac{2}{3}$
14. $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$
15. e.g. $\frac{2}{4}, \frac{3}{6}, \frac{4}{8}$
16. $\frac{3}{8}$
17. Reena 5, Kiran 6, left 4
18. 30 play; 10 do not
19. 24 sold; 36 remain
20. $\frac{1}{3}$; 8 biscuits each

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 3: Angles as Turns – Answer Key

1. Straight angle (180°)
2. 90°
3. Straight angle
4. 360°
5. e.g. a book, a window, a table corner

6. Right angle
7. Right angle (90°)
8. Right angle
9. 540°
10. East
11. 30° acute, 90° right, 120° obtuse, 180° straight
12. 180°
13. $\frac{1}{3}$ turn = 120°
14. Obtuse
15. 270°
16. Door B is obtuse; difference is 60°
17. 720°
18. 90° (a right angle)
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 4: We the Travellers - II – Answer Key

1. 105 minutes
2. 40 minutes
3. 11:00 am
4. 2 hours 30 minutes
5. 2:00 pm
6. 180 km
7. Bus B; 1h50min vs 2h15min, shorter by 25 minutes
8. 60 km/h
9. 37 minutes
10. 6:00 pm
11. Halt 15 min; total time 6h15min

12. 600 km/h
13. Friend B (12 km/h vs 10 km/h)
14. 7:50 am
15. 6:25 pm
16. $5\text{h} + 20\text{min} + 2\text{h} = 7\text{h } 20\text{min}$
17. Bus 3, then Bus 1, then Bus 2
18. Old: 80 km, New: 120 km, extra 40 km
19. Teacher Verification Required
20. 3 hours; arrive at 9:00 am

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 5: Far and Near – Answer Key

1. 10 mm
2. 1,000 m
3. 2.5 m
4. 3,000 m
5. m (metres)
6. 425 cm
7. 3.5 km
8. 1 m 42 cm
9. 120 cm
10. 3 m, 350 cm, 0.3 km, 3,500 mm
11. Asha's; longer by 5 cm
12. 4,100 m
13. 125 cm
14. 2,500 m
15. No, since $350\text{ cm} = 3.5\text{ m}$, which is taller than 3.2 m
16. 5.05 km
17. 8 pieces; nothing left over ($8 \times 125\text{ cm} = 1,000\text{ cm}$)

18. 800 m
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 6: The Dairy Farm – Answer Key

1. 1,000 ml
2. 2,500 ml
3. 3 l 750 ml
4. 1 litre bottle
5. 4,000 ml
6. 15 l
7. 5 l 400 ml
8. 30 packets
9. 45 l
10. 36 l 750 ml
11. 40 cups
12. 1,950 l
13. 315 l
14. 1 l 450 ml
15. 4.5 l
16. 680 l
17. Total = 57 l 50 ml; yes, it fits (less than 60 l)
18. Buffalo: 9 l 500 ml each, $\times 4 = 38$ l; Cow: 7 l 500 ml each, $\times 6 = 45$ l; total = 83 l
19. Teacher Verification Required
20. Rs. 784

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 7: Shapes and Patterns – Answer Key

1. 6
2. Sphere
3. 6
4. Circle
5. 3
6. 1 curved surface, 1 flat face
7. 100 m
8. 32 cm
9. 10, 12
10. 6 faces (all rectangles)
11. 260 m
12. Blue
13. 9, 11
14. 45 cm
15. 12
16. 12 m
17. Triangle
18. 30 cm
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 8: Weight and Capacity – Answer Key

1. 1,000 g
2. 2,300 g
3. 4 kg 500 g
4. 1 kg
5. 3,000 g

6. 6 kg 250 g
7. 4 kg 250 g
8. 2 kg 350 g
9. 25 packets
10. 7 kg 500 g
11. 8 kg 250 g
12. 324 kg 400 g
13. 3 kg 600 g
14. 1 kg 500 g
15. 12 kg 500 g
16. 16 bags; 1 kg left over
17. 36 packs
18. Total 74 kg 500 g; average $\approx$ 24 kg 833 g
19. Teacher Verification Required
20. 4 kg 250 g

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 9: Coconut Farm – Answer Key

1. 144
2. 12
3. 72
4. 12
5. 140
6. 480
7. 10
8. 12
9. 700
10. Rs. 600
11. 270 trees; $270 \times 60 = 16,200$ coconuts

12. Rs. 1,500
13. 1,000 sq m
14. 15
15. 140
16. About 12 trees
17. Cost Rs. 20,000; Revenue Rs. 27,300; Profit Rs. 7,300
18. 60 trees
19. Teacher Verification Required
20. 12 coconuts

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 10: Symmetrical Designs – Answer Key

1. Infinitely many
2. Yes, a vertical line through the middle
3. 3
4. No
5. e.g. A, M, T, U, W, Y
6. 2
7. A mirror image of the drawn half
8. Yes, 5 lines
9. Square
10. Teacher Verification Required
11. 6
12. Draw the mirror image of the left half on the right side
13. 6
14. Hexagon (6 vs 5)
15. Yes, it has both a vertical and a horizontal line of symmetry
16. 3 red dots and 2 blue dots, mirrored
17. 20 times

18. Line symmetry, because the fold acts as the line of symmetry
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 11: Grandmother's Quilt – Answer Key

1. $\frac{1}{2}$
2. $\frac{1}{4}$
3. 64 sq cm
4. 60 sq cm
5. 4
6. Side 45 cm; area 2,025 sq cm
7. $\frac{3}{8}$
8. 12 cm
9. 400 sq cm
10. 30
11. 2,000 sq cm
12. $\frac{1}{2}$
13. 300 cm = 3 m
14. 60
15. 30 squares, 20 triangles
16. $\frac{1}{16}$
17. e.g. 10×6 and 12×5
18. 32 red, 32 white
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 12: Racing Seconds – Answer Key

1. 60
2. 120 seconds
3. 2 min 30 sec
4. 90 seconds (since 1 min 20 sec = 80 sec)
5. 195 seconds
6. Asha wins by 4 seconds
7. 180 seconds = 3 minutes
8. 150 seconds = 2 min 30 sec
9. 200 seconds = 3 min 20 sec
10. 68 seconds
11. 55, 58, 60, 62 sec
12. 40 seconds
13. Heat 2 winner; difference 0.4 sec
14. The first lap
15. 0.7 seconds
16. 50.5 seconds
17. 0.5 seconds
18. 2 hours 27 min 20 sec
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 13: Animal Jumps – Answer Key

1. 1.5 m
2. 2.75 m
3. 4 m
4. 0.5, 0.8, 1.05, 1.2 m
5. 2.00 m

6. 0.75 m
7. 2.5 m
8. 0.2 m
9. 3.0 m
10. 0.75 m
11. Squirrel, cat, dog, monkey
12. 15 m
13. 4.8 m
14. 0.35 m
15. 2.70 m
16. Total 8.5 m; average ≈ 2.83 m
17. Athlete: 2.0 m; third animal: 1.8 m
18. 3 leaps
19. Teacher Verification Required
20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 14: Maps and Locations – Answer Key

1. North, South, East, West
2. Clockwise
3. North
4. East
5. North
6. 40 km
7. South
8. East-West
9. 6 km East, 4 km North
10. 200 km
11. Walk 2 km West and 1 km South

12. 1 hour
13. 350 m
14. 5 cm
15. East
16. Net North = 4 km (actual), East = 6 km (actual)
17. 30 km East, 20 km North, 30 km West; total 80 km
18. 5.5 cm
19. Teacher Verification Required
20. The scale, and the start and end points

Reflection & Real-Life Questions: Teacher Verification Required.

Worksheet 15: Data Through Pictures – Answer Key

1. 30 books
2. It tells what each symbol represents (its value)
3. The value/quantity of that category
4. 25
5. It is easier and quicker to compare data visually
6. 12
7. Cricket
8. 10
9. 12
10. March
11. 95 mm
12. 100 mangoes
13. Thursday, 10 mangoes
14. Teacher Verification Required
15. $\frac{15}{40} = \frac{3}{8}$
16. Asha, 15 books
17. 22

18. Rs. 4,400

19. Teacher Verification Required

20. Teacher Verification Required

Reflection & Real-Life Questions: Teacher Verification Required.